

CAR T-cell therapy

CART T-cell therapy (or CAR-T therapy) is a type of immunotherapy being used to treat some children and young adults with acute lymphoblastic leukaemia and some adults with non-Hodgkin lymphoma.

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What is CAR-T therapy?

CAR-T stands for chimeric antigen receptor T-cell. This therapy is a type of [immunotherapy](#). It works by changing some of your body's immune cells to make them better at fighting cancer cells. Another name for CAR-T therapy is adoptive cell therapy.

CAR-T therapy is being used to treat some children and young adults with [acute lymphoblastic leukaemia](#) and some adults with [non-Hodgkin lymphoma](#).

[Clinical trials](#) are trying to find out if it might work against other types of cancer.

CAR-T therapy and the immune system

The immune system helps protect us from [infection](#) or abnormal cells in our body (such as cancer cells).

White blood cells called T-cells are an important part of the immune system. They move around the body looking for cells that may be harmful. When T-cells find abnormal cells, they attack them and release chemical messengers (cytokines). These cytokines bring more immune cells to the area to kill the abnormal cells.

The immune system does not always recognise cancer cells as abnormal. This lets the cancer grow undetected. CAR-T therapy changes T-cells so that they recognise cancer cells and destroy them.

Having CAR-T therapy

CAR-T therapy involves 5 stages. These take place over a few weeks.

Stage 1



T-cells are taken from your blood. This takes about 3 hours. A nurse puts a short, thin tube ([cannula](#)) into a vein in each arm. Each cannula is connected by tubes to a machine called a cell separator. Blood goes from one arm into the cell separator, which removes the T-cells from your blood. The remaining blood and blood cells are returned to you through the cannula in your other arm.

Stage 2



The T-cells that have been removed are sent to a laboratory to be prepared. This takes 2 to 3 weeks. Scientists change the T-cells so that they recognise the cancer. These altered cells are called CAR T-cells. The number of CAR T-cells is then increased (expanded) in the lab.

Stage 3



You have [chemotherapy](#) to prepare your body to accept the CAR T-cells before they are given to you.

Stage 4



The CAR T-cells are given back into your bloodstream, usually through a drip but sometimes by injection. Once in your body, the T cells will begin to attack the cancer.

Stage 5



You are closely monitored for side effects and given treatment for these as needed.

When is CAR-T therapy used?

There are two types of CAR-T therapy licensed for use:

- tisagenlecleucel (Kymriah®)

- [axicabtagene ciloleucel \(Yescarta®\)](#).

In the UK they may be used to treat:

- children and young adults with B-cell acute lymphoblastic leukaemia (ALL), whose first treatment has not worked.
- adults with [diffuse large B-cell lymphoma](#) or [primary mediastinal large B-cell lymphoma](#) that is still growing after two or more treatments.

In the UK, an estimated 200 people have these treatments each year. It is only available in a small number of specialist hospitals. There is a risk of severe side effects, so CAR-T therapy is not suitable for everyone.

CAR-T therapy is also being tested in other cancers, mainly [blood cancers](#). Some people may be offered it as part of a [clinical trial](#).

Side effects of CAR-T therapy

You will have CAR-T therapy during a stay in hospital. The cancer team will monitor you closely for any side effects. CAR-T therapy is a new treatment. This means doctors do not know about all the side effects or any longer-term risks. After you leave, you will be asked to stay close to the hospital for the first few weeks.

When you are ready to go home, your healthcare team will tell you about side effects you may have. They will also give you a 24-hour phone number to contact them if you have side effects or feel unwell. This means you can be treated straight away if needed.

We have listed the most common side effects below.

Cytokine Release Syndrome (CRS)

This side effect usually happens within the first 10 days of treatment. Cytokines are chemical messengers that increase the effects of the immune system. CAR-T therapy causes the body to release large amounts of cytokines. This can cause flu-like symptoms, which can range from mild to severe. Some people may be monitored and treated in intensive care until this improves.

Symptoms include:

- a high temperature
- a fast heart rate
- low blood pressure and dizziness
- muscle and joint pain
- diarrhoea
- feeling sick and being sick
- difficulty breathing.

There are treatments that can help manage these side effects.

Effects on the brain (neurological side effects)

Sometimes CAR T-cells can affect the brain. Doctors call this neurotoxicity. It can happen within the first 4 to 8 weeks of treatment. Symptoms can range from mild to severe. Symptoms might include:

- headaches
- reduced consciousness (being less alert)
- confusion
- difficulty speaking
- loss of balance
- seizures (fits).

These symptoms usually get better within 1 to 2 weeks. If they are severe, there are treatments that which can help. Some people are monitored in intensive care until the symptoms improve.

Effects on blood cells

CAR-T therapy can reduce the levels of all of your blood cells. If you have a low number of white blood cells, this will increase your risk of [infections](#). The cancer team will give you advice on how to protect yourself from infection.

Blood cell numbers usually recover slowly after treatment. After treatment with tisagenlecleucel or [axicabtagene ciloleucel](#), the levels of white blood cells called B-cells may stay low for longer. And it is possible they may not fully recover.

Your cancer specialist can tell you more about CAR-T therapy.

About our information

References



Below is a sample of the sources used in our CAR-T therapy information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk.

Steven Feins et al. CRITICAL REVIEW. An introduction to chimeric antigen receptor (CAR) T-cell immunotherapy for human cancer. American Journal of Hematology. 2019.

NICE (National Institute for Health and Care Excellence). Tisagenlecleucel for treating relapsed or refractory diffuse large B-cell lymphoma after 2 or more systemic therapies. Technology appraisal guidance [TA567]. 2019.

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Anne Parker](#), Consultant Haematologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

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⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help